

E3/DUB specificity prediction and discovery workflow

Raw data

UbiBrowser
BioGRID
UniProt
TISCH2

Data curation

< Positive pairs
Negative sampling
PPI / localization / scRNA filters

Protein representation

< ESM2 / ProtBERT
Per-sequence
Per-residue

Pair-level learning

< XGBoost
Neural scorer
Fusion

Graph learning

< GraphSAGE
HeteroGNN
HGT

Discovery

< Novel predictions
Explainability
Cancer prioritization